

CONSORT AGDM, AB, Canada

WMO: 712230

Lat: 51.9369N

Lon: 110.7131W

Elev: 2533

StdP: 13.40

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.1	-17.1	-29.0	1.2	-21.8	-23.7	1.6	-17.1	31.3	21.7	27.3	20.4	9.2	290	0.743

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	86.0	62.2	82.1	61.6	78.3	60.0	65.8	78.9	63.9	76.9	62.1	74.4	11.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.3	88.8	71.0	59.2	82.4	69.4	57.3	76.9	67.0	32.1	79.3	30.5	77.2	29.1	74.2	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.2	24.7	21.7	DB	-29.3	92.2	5.5	4.2	-33.3	95.2	-36.5	97.6	-39.6	99.9	-43.6	103.0	(4)
(5)			WB	-29.7	69.8	5.4	2.3	-33.5	71.4	-36.7	72.8	-39.7	74.1	-43.7	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	11.8	11.7	24.4	39.2	51.1	58.1	64.0	61.9	52.7	39.7	23.1	12.3		
	DBStd	22.13	15.29	13.76	13.81	9.57	7.79	5.68	5.59	6.45	8.63	9.67	12.85	14.37		
	HDD50	5856	1183	1072	794	338	82	6	0	2	67	337	807	1168		
	HDD65	10098	1648	1492	1258	775	432	217	87	138	378	783	1257	1633		
	CDD50	1350	0	0	0	13	116	248	433	371	149	19	1	0		
	CDD65	115	0	0	0	0	2	9	54	41	9	0	0	0		
	CDH74	1933	0	0	1	4	89	176	773	646	234	10	0	0		
	CDH80	560	0	0	0	0	10	33	246	196	75	0	0	0		
(14) Wind	WSAvg	10.8	11.6	10.7	11.1	11.8	11.3	10.7	9.6	9.4	10.1	11.1	11.3	11.0		
(15) Precipitation	PrecAvg	13.0	0.6	0.3	0.5	0.8	1.5	2.9	2.1	1.7	1.0	0.6	0.5	0.4		
	PrecMax	18.5	1.9	0.7	1.4	2.4	3.1	5.9	5.0	3.3	3.3	1.3	1.7	0.7		
	PrecMin	6.8	0.1	0.0	0.1	0.1	0.1	0.7	0.8	0.2	0.0	0.2	0.0	0.1		
	PrecStd	3.0	0.4	0.2	0.3	0.6	0.9	1.3	0.9	0.8	0.8	0.3	0.4	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	42.3	58.3	72.5	81.2	84.4	91.2	90.8	87.7	74.7	56.7	43.6		
		MCWB	36.0	36.0	45.5	51.3	56.3	61.3	64.1	62.6	59.6	53.8	43.8	37.6		
	2%	DB	38.7	37.6	51.7	66.7	76.3	78.8	86.2	85.1	80.9	66.0	48.3	39.2		
		MCWB	34.3	33.3	41.2	48.4	54.1	60.0	64.5	61.3	57.6	50.3	39.7	34.9		
	5%	DB	36.1	34.2	46.1	61.1	72.0	75.0	82.1	81.0	74.6	60.0	42.9	35.7		
		MCWB	32.5	30.9	38.5	45.1	52.8	58.6	63.4	61.1	55.7	47.2	36.5	31.8		
	10%	DB	32.9	30.9	40.7	56.0	67.5	71.4	77.9	77.0	69.2	54.7	38.8	31.5		
		MCWB	30.0	28.4	35.5	42.7	50.6	56.6	62.1	60.0	53.5	44.2	34.0	28.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	36.9	46.7	52.1	58.4	65.5	70.2	67.1	61.6	55.3	44.8	38.0		
		MCDB	40.7	41.7	57.8	70.2	75.0	77.5	83.1	79.4	80.2	71.6	54.6	42.5		
	2%	WB	34.7	33.7	41.8	48.9	55.9	62.4	66.8	64.8	58.8	51.3	40.4	35.0		
		MCDB	38.3	37.3	50.4	65.1	72.1	74.5	79.8	78.5	77.0	63.3	47.8	39.0		
	5%	WB	32.7	31.2	38.6	46.0	53.9	60.1	64.9	62.8	56.6	48.0	36.9	31.9		
		MCDB	35.9	34.1	45.1	58.8	69.2	71.8	77.9	76.4	72.3	58.6	41.6	35.5		
	10%	WB	30.1	28.3	35.6	43.4	51.8	58.1	63.2	60.9	54.3	45.2	34.3	28.8		
		MCDB	32.9	30.7	40.6	54.3	65.6	68.9	75.2	73.9	67.8	54.2	38.2	31.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	17.0	17.7	21.4	23.1	20.5	23.2	23.6	23.4	20.6	16.4	15.6		
		MCDBR	18.0	19.4	24.2	31.1	30.2	27.1	30.2	31.4	33.4	29.7	21.8	18.7		
	5% WB	MCWBR	15.0	16.1	15.2	15.6	13.6	12.6	12.5	12.6	14.8	16.0	14.6	14.7		
		MCWBR	18.3	19.0	22.8	28.6	26.7	23.9	26.2	27.1	30.7	28.4	20.3	18.9		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.304	0.291	0.340	0.366	0.370	0.362	0.377	0.322	0.284	0.259	0.250		
	taud		2.341	2.225	2.439	2.326	2.301	2.335	2.396	2.321	2.454	2.538	2.469	2.340		
	Ebn at Noon		244	256	290	285	281	280	280	269	275	268	245	226		
	Edh at Noon		19	29	28	36	39	38	35	36	28	21	16	16		
(44) All-Sky Solar Radiation	RadAvg		362	698	1144	1526	1846	1910	2037	1657	1182	712	371	272		
	RadStd		33	57	86	133	124	123	92	128	113	91	34	25		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page